Ashwin K M

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Papers

- **Learn–Master–Teach Tuning: From Pupil to Professor**
  A Human-Pedagogy Inspired Fine-Tuning Paradigm for LLMs
- **FOCAL: Decoupling Conflicting Objectives in LLM Domain Adaptation via Temporal Orthogonalization**

Education

- **Indian Institute of Science, Bangalore** TGPA: 8.3
  B.Tech in Mathematics and Computing 2023 – 2027

Research Experience

- **Zenteiq AI**
  ML Research Engineer May 2025 – Dec 2025
  - Investigated whether training LLMs through a human-inspired **Learn–Master–Teach** lifecycle improves mathematical reasoning accuracy beyond static fine-tuning.
  - Designed a multi-stage training pipeline combining curriculum-grounded learning, interactive concept refinement, feedback-driven practice, multi-teacher distillation, self-reflection, hallucination mitigation, and pedagogical reinforcement learning to systematically improve reasoning reliability.
  - Found that the integrated pipeline yields **+10–17 pp accuracy gains** over the base model on CBSE-style math benchmarks, while revealing that feedback-driven refinement and teaching-oriented objectives are critical for stable gains beyond knowledge injection alone.
- **QUEST Lab**
  Research Intern Aug 2026 – Dec 2026
  - Investigating how **multimodal memory** can help turbine digital twins relate sensor histories to turbine schematics, manuals, and earlier operating experience, with the aim of improving diagnostic reasoning as operating conditions and equipment behavior change.
  - Designing **context-sensitive retrieval and consolidation** across relational, visual, and textual representations to preserve equipment relationships, temporal context, and uncertainty. The research examines how a twin can retain informative rare events and revise earlier knowledge as operating conditions change.
  - Developed turbine visualization and an AI copilot as the experimental foundation for **Quest TwinOps**. Current research focuses on a feedback loop that compares explanations, identifies informative measurements or approved tests, and uses subsequent observations to revise the twin's memory.

Selected Projects

- **MambaRecall: Coarse-to-Fine Token Memory** Aug 2026 – Dec 2026
  - Investigated how recurrent language models retain information and retrieve details needed for later predictions. Analyzed how fixed-state compression, segment summaries, and averaging in Hierarchical Memory Mamba (HMM) can obscure those details. This motivated a memory architecture that separates finding relevant history from reading its contents.
  - Developed a **selective token-memory adapter** for frozen Mamba2 that combines a dedicated sliding-window path with selective attention over encoded tokens in distant chunks. Adapted RAMba's hierarchical sparse attention to separate coarse chunk selection from fine-grained reads, and used selected relevance scores in differentiable fusion weights to connect memory selection to prediction loss.
  - Evaluated the complete adapter against local HMM on PG-19 using Mamba2-1.3B across three paired seeds. Reduced mean 32K perplexity from 13.384 to 13.006 (**2.82%**), with improvements at all four evaluated context lengths from 4K to 32K while retaining the pretrained recurrent backbone.
- **FOCAL** Jan 2026 – Apr 2026
  - Current AI scientists fail to identify **meaningful research directions** for scientific discovery in domains with limited prior work as they don't build foundational understanding nor iteratively refine their hypothesis.
  - Proposed a **multi-agent framework** that acquires prerequisite knowledge, generates and critiques research hypotheses, and designs experiments to evaluate promising research directions.
  - Demonstrated up to **13.2 percentage-point improvements** over ABMLL across common-sense, mathematical, coding, and medical reasoning benchmarks, while exhibiting stronger out-of-domain generalization than competing meta-learning approaches.

- **Latent Diffusion Research Agents** Jan 2026 – May 2026
  - Current deep research agents rely on **repeated text-mediated communication** between agents, introducing cumulative information loss, vocabulary bottlenecks, and increasing inference costs across long-horizon research workflows.
  - Developed a latent-space collaboration framework that replaces internal text exchanges with **KV-cache transfer**, localizes decoding to tool-interaction boundaries, and provides theoretical guarantees on convergence, information preservation, and iterative agent refinement.
  - Matched the performance of text-mediated baselines while reducing generated tokens by **33%** and wall-clock execution time by **28%**, validating latent communication as a more efficient mechanism for multi-agent coordination.
- **Benchmarking Offline Reinforcement Learning Algorithms** Jan 2025 – Apr 2025
  - Investigated the robustness of standard offline reinforcement learning algorithms (Decision Transformer, Implicit Q-Learning, TD3-BC, AWAC, ReBRAC, Behavior Cloning) by benchmarking them on MuJoCo continuous-control tasks using the **Minari dataset format**, motivated by the need to move beyond legacy D4RL pipelines and evaluate algorithms under modern, reproducible offline-RL data interfaces.
  - Refactored and rebuilt the offline RL training and evaluation pipeline to support **Minari + MuJoCo environments**, addressing incompatibilities with D4RL assumptions (dataset APIs, environment registration, trajectory handling), and ensuring fair, like-for-like comparisons across algorithms under a unified experimental setup.
  - Conducted an architectural ablation replacing standard `nn.Linear` layers with `NdLinear` layers across policies and critics, observing marginal performance differences in single-dimensional settings but identifying a key limitation of the benchmark—namely that NdLinear's advantages are likely to emerge in higher-dimensional tensor structures—highlighting a concrete direction for future multi-modal or structured-state offline RL evaluation.
- **IssueOps: Intelligent Repository Management** Aug 2025 – Dec 2025
  - Investigated whether a **multi-agent, event-driven automation pipeline** can reduce issue triage latency and maintainer burnout in large GitHub repositories, motivated by evidence that developers lose **~7.3 hours/week** to triage and face **3+ day delays** before issues are categorized or routed in high-noise repositories.
  - Designed and implemented a **fully serverless, asynchronous architecture** (AWS Lambda, EventBridge, DynamoDB) comprising *Intake, Triage, Assignment, and SLA agents*; integrated **LLM-based classification** (Llama-3 for spam filtering and metadata cleanup, Mistral-7B for issue type/priority/component inference) and **availability-aware routing** by inferring contributor expertise from commit history and working hours from activity timestamps.
  - Validated **end-to-end automation on live repositories**, processing batches of issues in **2-5 minutes per repository** and operating in parallel across repos; peer testing revealed accuracy limitations of a tuned BERT classifier despite >85% test accuracy, motivating a switch to **Mistral-7B via AWS Bedrock**, which improved real-world labeling reliability and highlighted trade-offs around cold starts and per-issue inference cost.

Skills

- **Programming & Core ML:** Python, PyTorch, HuggingFace, LLM Routing, Multi-Fidelity Systems
- **LLMs & Agentic Systems:** PEFT, LangChain, LangGraph, vLLM, Unsloth
- **Reinforcement Learning:** Multi-Armed Bandits, TRL, OpenRLHF, Gymnasium, MuJoCo
- **Systems & Infrastructure:** Efficient Inference, AWS Cloud, Ray, Docker, Git, Weights & Biases

Achievements

- **World Scholar's Cup, Global Round (Manila)** - Placed **30th out of 1,000 teams** worldwide; won **8 Gold and 5 Silver medals**; qualified for the **Tournament of Champions at Yale University** 2019
- **National Talent Search Examination (NTSE)** - Stage I Awardee with **Rank 55** 2021
- **Kishore Vaigyanik Protsahan Yojana (KVPY)** - Secured **Top 2000 rank** in SA stream 2021

Positions of Responsibility

Core Coordinator — Nexus (SciTech Fest), Rhapsody 2025, IISc Dec 2024 – Mar 2025

Founded **Nexus** at IISc, organizing AI and robotics competitions and building partnerships with Autodesk, L&T, OLA, and ART-PARK.

Sponsorship Coordinator — Pravega XI, IISc Aug 2024 – Dec 2024

Secured sponsorships and industry partnerships through corporate outreach for Pravega's science and technology programme.

Event Coordinator — RevCoding, Pravega 2025, IISc Jul 2024 – Jan 2025

Created **RevCoding**, a reverse-coding competition combining hidden clues, algorithmic reasoning, and narrative puzzles across multiple stages.